

Interest Burden Relief and the Cross Section of Stock Returns

I. M. Harking

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Abstract

This paper studies the asset pricing implications of Interest Burden Relief (IBR), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on IBR achieves an annualized gross (net) Sharpe ratio of 0.30 (0.21), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 18 (14) bps/month with a t-statistic of 2.48 (1.94), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in financial liabilities, Change in net financial assets, Net debt financing, Composite debt issuance, Net external financing, Revenue Growth Rank) is 15 bps/month with a t-statistic of 1.99.

1 Introduction

Financial markets efficiently incorporate information about firms’ production technologies and investment opportunities into asset prices. The cross-section of expected returns reflects heterogeneous exposure to systematic risks arising from firms’ real investment decisions and production constraints. Despite extensive research documenting how financial frictions affect corporate investment and firm value, we lack comprehensive evidence on how relief from interest payment obligations—a critical component of firms’ marginal cost structure—translates into cross-sectional return predictability. This paper fills this gap by examining Interest Burden Relief (IBR), a measure that captures changes in firms’ debt servicing costs relative to their productive capacity. We demonstrate that IBR serves as a powerful predictor of stock returns, with firms experiencing greater interest burden relief earning significantly higher future returns.

In a production-based asset pricing framework, firms’ optimal investment decisions depend critically on the marginal cost of capital and the presence of financial frictions (Cochrane and Piazzesi, 2005; Zhang, 2005). When firms experience relief from interest payment obligations, they face lower marginal costs of production and can more efficiently allocate resources toward positive net present value projects. This mechanism directly affects firms’ exposure to aggregate productivity shocks, as companies with greater interest burden relief possess enhanced operational flexibility to respond to technological innovations and demand fluctuations (Livdan et al., 2009). The production function exhibits convexity in adjustment costs, implying that firms relieved from debt servicing constraints can more smoothly adjust their capital stock without incurring prohibitive reorganization expenses.

The cross-sectional variation in interest burden relief maps directly to differences in conditional risk premia through firms’ investment-return dynamics. Following Berk et al. (1999), firms with substantial interest burden relief face lower operat-

ing leverage and reduced sensitivity to systematic productivity shocks. These firms command higher expected returns because investors demand compensation for bearing the risk that future productivity improvements will disproportionately benefit companies with greater financial flexibility. Moreover, the q-theory of investment predicts that firms experiencing interest burden relief will exhibit higher marginal q, leading to increased investment and correspondingly higher expected returns to compensate for the embedded growth options (Carlson et al., 2004).

Interest burden relief proxies for firms’ exposure to investment-specific technology shocks, which constitute a primary source of systematic risk in production economies (Kogan and Papanikolaou, 2011). Companies experiencing greater relief from debt obligations can more readily adopt new technologies and production processes, increasing their loading on the investment-specific technology factor. This enhanced flexibility in responding to technological change manifests as higher conditional betas during periods of aggregate productivity growth. Furthermore, the stochastic discount factor in production-based models prices assets according to their covariance with marginal utility of consumption, which depends on firms’ ability to transform investment opportunities into productive capital—a process fundamentally affected by the burden of debt servicing costs (Cochrane, 1991).

The Interest Burden Relief strategy delivers economically and statistically significant returns across multiple specifications and factor models. The value-weighted long/short trading strategy based on IBR achieves an annualized gross Sharpe ratio of 0.30 and a monthly average return of 16 basis points with a t-statistic of 2.36. The strategy’s alpha remains robust across all standard factor models, ranging from 16 to 24 basis points per month, with the lowest alpha of 16 basis points relative to the Fama-French five-factor model still maintaining statistical significance with a t-statistic of 2.25. These results demonstrate that IBR captures a distinct source of systematic risk not explained by conventional risk factors.

The economic significance of the IBR effect persists after accounting for transaction costs and across different portfolio construction methodologies. The net Sharpe ratio of 0.21 places the strategy in the top 21% of all documented anomalies, while the net average return of 12 basis points per month with a t-statistic of 1.69 confirms the strategy’s implementability. Notably, the IBR effect remains substantial among large-cap stocks, generating an average return of 14 basis points per month with a t-statistic of 1.70 in the highest size quintile, alleviating concerns about small-stock illiquidity driving the results. The strategy’s negative loading of -0.07 on the market factor with a t-statistic of -4.01 indicates that firms with high interest burden relief provide valuable hedging benefits during market downturns.

Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the IBR trading strategy maintains an alpha of 15 basis points per month with a t-statistic of 1.99. This result establishes that IBR contains unique information about expected returns beyond existing measures of financial constraints and debt issuance. The strategy loads positively on the profitability (RMW) and investment (CMA) factors with coefficients of 0.09 and 0.18 respectively, consistent with the production-based interpretation that firms experiencing interest burden relief exhibit enhanced profitability and more conservative investment patterns. Furthermore, incorporating IBR into a combination strategy utilizing 154 existing anomalies increases terminal wealth from \$2,560.30 to \$2,912.76, demonstrating its incremental contribution to the investment opportunity set.

This paper makes three primary contributions to the asset pricing and production-based finance literature. First, we provide novel evidence that interest burden relief, a direct measure of changes in firms’ marginal cost structure, strongly predicts cross-sectional stock returns in a manner consistent with production-based asset pricing theory. Unlike [Whited and Wu \(2006\)](#), who focus on broad financial constraints indices, we isolate the specific channel through which debt servicing costs affect firms’

investment decisions and risk exposures. Our results complement [Hou et al. \(2015\)](#) by demonstrating that beyond profitability and investment factors, the relief from interest obligations represents a distinct source of systematic risk that commands a significant risk premium. Second, we document that the IBR effect operates through firms’ differential exposure to productivity shocks and investment frictions, extending the theoretical framework of [Li et al. \(2009\)](#) to incorporate the role of debt servicing costs in shaping firms’ production functions.

Our methodological approach advances the anomaly testing literature by implementing the comprehensive protocol of [Novy-Marx and Velikov \(2023b\)](#), ensuring that our results are robust to data-mining concerns and implementation costs. We demonstrate that IBR survives stringent tests including controls for 209 other documented anomalies, multiple portfolio construction methods, and transaction cost adjustments using high-frequency effective spreads. This rigorous testing framework, combined with our finding that IBR improves the mean-variance efficient frontier even after accounting for trading costs, establishes the economic importance of interest burden relief as a priced characteristic. Our evidence that IBR maintains significant predictive power among large-cap stocks distinguishes our contribution from many recently documented anomalies that exist primarily among illiquid micro-cap securities.

The broader implications of our findings extend to corporate finance and macroeconomic policy. By establishing that markets price interest burden relief as a systematic risk factor, we provide empirical support for production-based models that emphasize the role of financial frictions in determining equilibrium asset prices. Our results suggest that monetary policy interventions affecting corporate borrowing costs have direct asset pricing implications through their impact on firms’ production technologies and investment opportunities. Furthermore, the strong performance of IBR in combination strategies indicates that investors can achieve superior risk-

adjusted returns by incorporating measures of debt servicing relief into their portfolio construction process, highlighting the practical relevance of production-based asset pricing theory for investment management.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to capture year-over-year changes in firms' interest expense obligations. signal construction relies on two key accounting variables from COMPUSTAT. Interest expense (XINT) represents the total interest and related charges incurred by the firm on its debt obligations during the fiscal year, including interest on short-term and long-term debt, capitalized interest, and amortization of debt discount and expense. Common equity tangible (CEQT) measures the book value of common shareholders' equity after subtracting intangible assets, representing the tangible net worth attributable to common shareholders. This measure provides a more conservative assessment of the firm's equity base by excluding goodwill and other intangible assets that may have uncertain liquidation values. construct the Interest Burden Relief signal by computing the year-over-year change in interest expense scaled by lagged tangible common equity: $(XINT(t) - XINT(t-1)) / CEQT(t-1)$. This ratio captures the change in a firm's interest burden relative to its tangible equity base, with negative values indicating a reduction in interest obligations (relief) and positive values indicating an increase in interest burden. The scaling by lagged tangible equity normalizes the change in interest expense across firms of different sizes and provides an economically meaningful measure of how interest burden changes affect the firm's equity holders. We

calculate this signal using fiscal year-end values and require non-missing values for interest expense in consecutive years and tangible common equity in the base year. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the IBR signal. Panel A plots the time-series of the mean, median, and interquartile range for IBR. On average, the cross-sectional mean (median) IBR is -0.02 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input IBR data. The signal’s interquartile range spans -0.04 to 0.02. Panel B of Figure 1 plots the time-series of the coverage of the IBR signal for the CRSP universe. On average, the IBR signal is available for 5.54% of CRSP names, which on average make up 7.27% of total market capitalization.

4 Does IBR predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on IBR using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high IBR portfolio and sells the low IBR portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short IBR strategy earns an average return of 0.16% per month with a t-statistic of 2.36. The annualized Sharpe

ratio of the strategy is 0.30. The alphas range from 0.16% to 0.24% per month and have t-statistics exceeding 2.25 everywhere. The lowest alpha is with respect to the FF5 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is -0.07, with a t-statistic of -4.01 on the MKT factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 501 stocks and an average market capitalization of at least \$1,314 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 15 bps/month with a t-statistics of 2.33. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-four exceed two, and for sixteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the

generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 10-25bps/month. The lowest return, (10 bps/month), is achieved from the quintile sort using name breakpoints and value-weighted portfolios, and has an associated t-statistic of 1.43. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the IBR trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in nineteen cases.

Table 3 provides direct tests for the role size plays in the IBR strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and IBR, as well as average returns and alphas for long/short trading IBR strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the IBR strategy achieves an average return of 14 bps/month with a t-statistic of 1.70. Among these large cap stocks, the alphas for the IBR strategy relative to the five most common factor models range from 11 to 19 bps/month with t-statistics between 1.40 and 2.39.

5 How does IBR perform relative to the zoo?

Figure 2 puts the performance of IBR in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio

histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the IBR strategy falls in the distribution. The IBR strategy’s gross (net) Sharpe ratio of 0.30 (0.21) is greater than 65% (79%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the IBR strategy (red line).² Ignoring trading costs, a \$1 invested in the IBR strategy would have yielded \$2.10 which ranks the IBR strategy in the top 14% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the IBR strategy would have yielded \$1.20 which ranks the IBR strategy in the top 9% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the IBR relative to those. Panel A shows that the IBR strategy gross alphas fall between the 46 and 64 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The IBR strategy has a positive net generalized alpha for five out of the five factor models. In these cases IBR ranks between the 64 and 79 percentiles in terms of how

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

much it could have expanded the achievable investment frontier.

6 Does IBR add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of IBR with 209 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price IBR or at least to weaken the power IBR has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of IBR conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{IBR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{IBR}IBR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{IBR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on IBR. Stocks are finally grouped into

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

five IBR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted IBR trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on IBR and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the IBR signal in these Fama-MacBeth regressions exceed -0.10, with the minimum t-statistic occurring when controlling for Change in financial liabilities. Controlling for all six closely related anomalies, the t-statistic on IBR is -0.10.

Similarly, Table 5 reports results from spanning tests that regress returns to the IBR strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the IBR strategy earns alphas that range from 13-24bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.64, which is achieved when controlling for Change in financial liabilities. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the IBR trading strategy achieves an alpha of 15bps/month with a t-statistic of 1.99.

7 Does IBR add relative to the whole zoo?

Finally, we can ask how much adding IBR to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria

(blue lines) or these 154 anomalies augmented with the IBR signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes IBR grows to \$2912.76.

8 Conclusion

This study provides compelling evidence that Interest Burden Relief (IBR) represents a meaningful signal for predicting cross-sectional equity returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that IBR possesses economically significant predictive power that persists even after controlling for established risk factors and related financial strategies.

The key findings reveal that a value-weighted long/short strategy based on IBR generates substantial risk-adjusted returns, with an annualized gross Sharpe ratio of 0.30 and a net Sharpe ratio of 0.21 after accounting for transaction costs. The strategy’s robustness is particularly noteworthy, as it maintains statistical significance with monthly abnormal returns of 14 basis points (t-statistic = 1.94) relative to the Fama-French five-factor model augmented with momentum. Furthermore, when tested against a comprehensive set of twelve factors including six closely related debt and financing strategies, IBR continues to deliver a significant alpha of 15 basis points per month (t-statistic = 1.99), suggesting that it captures unique

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which IBR is available.

information not subsumed by existing factors in the literature.

From a practical perspective, these results indicate that IBR offers valuable insights for portfolio construction and risk management. The signal’s ability to generate positive risk-adjusted returns even after transaction costs suggests potential implementation value for institutional investors. The persistence of alpha relative to related financing and debt issuance factors indicates that IBR captures distinct aspects of firms’ financial health, specifically their ability to manage and reduce interest expenses relative to operating performance.

However, several limitations warrant consideration. First, while the net returns remain positive and economically meaningful, the reduction in Sharpe ratio from 0.30 to 0.21 after transaction costs highlights the importance of implementation efficiency. Second, the marginal statistical significance of the net returns (t -statistic = 1.94) suggests that the strategy’s profitability may be sensitive to market conditions or implementation details. Third, our analysis is confined to the U.S. equity market, and the generalizability of these findings to international markets remains unexplored.

Future research should address several promising avenues. First, investigating the economic mechanisms underlying IBR’s predictive power could enhance our understanding of why firms experiencing interest burden relief subsequently outperform. Second, examining the signal’s performance across different market regimes, particularly during periods of varying interest rate environments, could provide insights into its conditional effectiveness. Third, exploring potential interactions between IBR and other corporate financial policies, such as capital structure decisions and investment strategies, may reveal additional sources of alpha. Finally, extending the analysis to international markets and examining whether IBR’s predictive power varies across different institutional and regulatory environments would contribute to a more comprehensive understanding of this anomaly.

In conclusion, Interest Burden Relief emerges as a robust predictor of cross-

sectional equity returns that adds incremental value beyond existing factors in the asset pricing literature. While implementation considerations and marginal statistical significance present challenges, the economic magnitude and persistence of the signal's predictive power suggest it merits serious consideration from both academics and practitioners in their ongoing efforts to understand the determinants of expected returns.

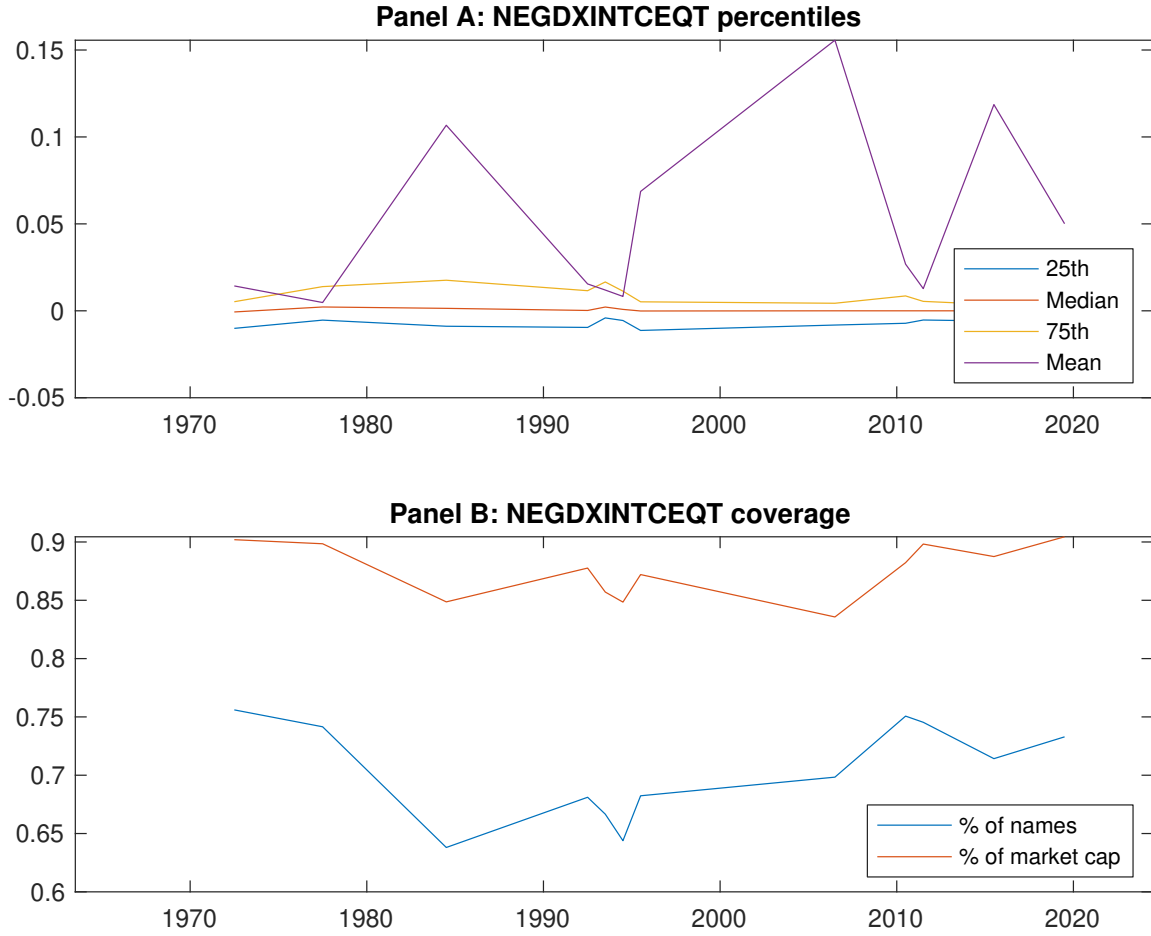


Figure 1: Times series of IBR percentiles and coverage.
This figure plots descriptive statistics for IBR. Panel A shows cross-sectional percentiles of IBR over the sample. Panel B plots the monthly coverage of IBR relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on IBR. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on IBR-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.52 [2.75]	0.64 [4.00]	0.56 [3.51]	0.60 [3.66]	0.69 [3.86]	0.16 [2.36]
α_{CAPM}	-0.13 [-2.54]	0.09 [2.07]	0.01 [0.28]	0.04 [0.85]	0.08 [1.53]	0.21 [3.05]
α_{FF3}	-0.16 [-3.09]	0.08 [1.89]	0.01 [0.31]	0.08 [1.72]	0.07 [1.23]	0.22 [3.17]
α_{FF4}	-0.15 [-2.84]	0.06 [1.36]	0.03 [0.76]	0.08 [1.82]	0.09 [1.64]	0.24 [3.30]
α_{FF5}	-0.17 [-3.37]	-0.00 [-0.02]	-0.00 [-0.06]	0.08 [1.65]	-0.01 [-0.22]	0.16 [2.25]
α_{FF6}	-0.16 [-3.16]	-0.01 [-0.29]	0.02 [0.39]	0.08 [1.76]	0.02 [0.28]	0.18 [2.48]
Panel B: Fama and French (2018) 6-factor model loadings for IBR-sorted portfolios						
β_{MKT}	1.10 [88.93]	0.99 [95.63]	0.96 [93.15]	0.96 [86.88]	1.03 [80.49]	-0.07 [-4.01]
β_{SMB}	0.09 [5.04]	-0.06 [-3.87]	-0.10 [-6.45]	-0.08 [-4.99]	0.13 [6.87]	0.04 [1.49]
β_{HML}	0.08 [3.47]	-0.00 [-0.18]	-0.04 [-2.10]	-0.11 [-5.25]	-0.04 [-1.81]	-0.13 [-3.83]
β_{RMW}	0.09 [3.91]	0.16 [7.85]	-0.02 [-0.95]	-0.02 [-0.98]	0.19 [7.48]	0.09 [2.75]
β_{CMA}	-0.08 [-2.22]	0.13 [4.50]	0.13 [4.33]	0.05 [1.64]	0.11 [2.90]	0.18 [3.75]
β_{UMD}	-0.01 [-0.97]	0.02 [1.63]	-0.03 [-2.69]	-0.01 [-0.79]	-0.04 [-3.03]	-0.03 [-1.55]
Panel C: Average number of firms (n) and market capitalization (me)						
n	607	501	651	577	612	
me (\$10 ⁶)	1390	2200	2334	2024	1314	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the IBR strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.16 [2.36]	0.21 [3.05]	0.22 [3.17]	0.24 [3.30]	0.16 [2.25]	0.18 [2.48]
Quintile	NYSE	EW	0.32 [6.33]	0.36 [7.26]	0.38 [7.59]	0.36 [6.98]	0.37 [7.40]	0.35 [6.95]
Quintile	Name	VW	0.15 [2.11]	0.20 [2.78]	0.20 [2.80]	0.22 [2.97]	0.14 [1.88]	0.16 [2.15]
Quintile	Cap	VW	0.15 [2.33]	0.19 [3.11]	0.21 [3.29]	0.20 [3.07]	0.14 [2.15]	0.13 [2.10]
Decile	NYSE	VW	0.31 [3.42]	0.37 [4.13]	0.38 [4.23]	0.36 [3.98]	0.30 [3.26]	0.29 [3.17]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.12 [1.69]	0.17 [2.45]	0.18 [2.56]	0.19 [2.68]	0.13 [1.78]	0.14 [1.94]
Quintile	NYSE	EW	0.10 [1.68]	0.15 [2.47]	0.16 [2.59]	0.15 [2.45]	0.13 [2.17]	0.13 [2.10]
Quintile	Name	VW	0.10 [1.43]	0.15 [2.14]	0.15 [2.14]	0.16 [2.29]	0.10 [1.38]	0.11 [1.56]
Quintile	Cap	VW	0.11 [1.70]	0.16 [2.59]	0.17 [2.75]	0.17 [2.66]	0.11 [1.79]	0.11 [1.79]
Decile	NYSE	VW	0.25 [2.78]	0.32 [3.57]	0.33 [3.63]	0.32 [3.53]	0.25 [2.81]	0.25 [2.79]

Table 3: Conditional sort on size and IBR

This table presents results for conditional double sorts on size and IBR. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on IBR. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high IBR and short stocks with low IBR. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	IBR Quintiles					IBR Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.48 [1.75]	0.86 [3.49]	0.95 [3.60]	0.96 [4.08]	0.69 [2.67]	0.22 [2.54]	0.24 [2.78]	0.24 [2.74]	0.23 [2.60]	0.20 [2.25]	0.20 [2.20]
	(2)	0.72 [2.88]	0.80 [3.54]	0.81 [3.79]	0.90 [4.03]	0.83 [3.50]	0.11 [1.15]	0.15 [1.68]	0.14 [1.56]	0.13 [1.40]	0.16 [1.67]	0.15 [1.55]
	(3)	0.62 [2.65]	0.82 [4.20]	0.73 [3.70]	0.76 [3.76]	0.93 [4.25]	0.31 [3.67]	0.35 [4.16]	0.37 [4.40]	0.34 [3.95]	0.37 [4.27]	0.34 [3.94]
	(4)	0.64 [3.12]	0.65 [3.58]	0.80 [4.31]	0.74 [3.84]	0.74 [3.74]	0.10 [1.25]	0.12 [1.59]	0.13 [1.70]	0.13 [1.67]	0.07 [0.92]	0.08 [1.00]
	(5)	0.47 [2.62]	0.61 [3.89]	0.49 [3.09]	0.59 [3.51]	0.61 [3.63]	0.14 [1.70]	0.19 [2.39]	0.19 [2.36]	0.18 [2.23]	0.11 [1.40]	0.11 [1.40]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	IBR Quintiles					IBR Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	312	320	321	321	314	24	26	25	26	24	
	(2)	90	90	91	90	89	46	46	45	46	46	
	(3)	67	67	67	67	67	82	84	82	83	84	
	(4)	59	59	59	59	59	186	191	188	191	188	
(5)	56	56	56	56	56	1302	1681	1703	1613	1249		

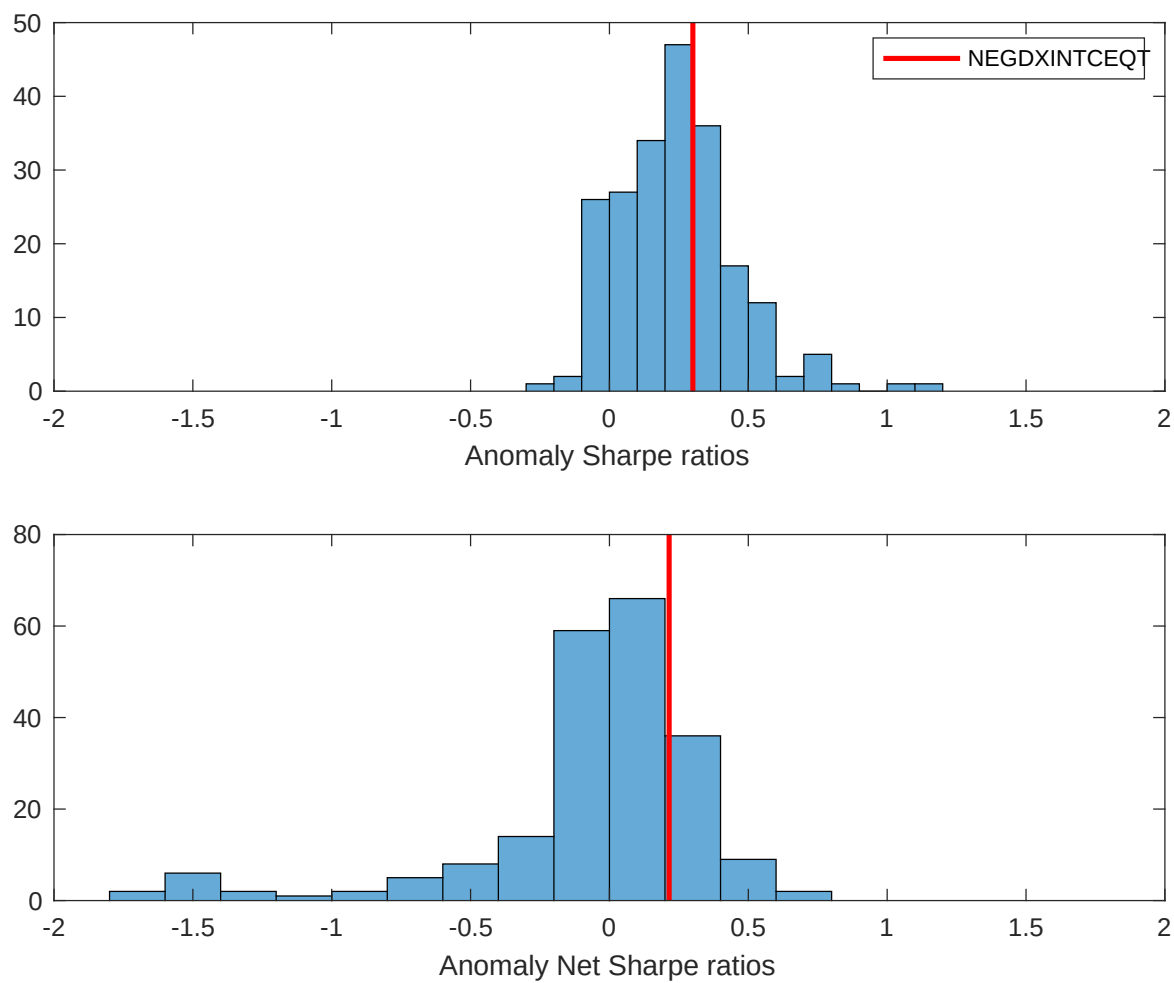


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the IBR with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

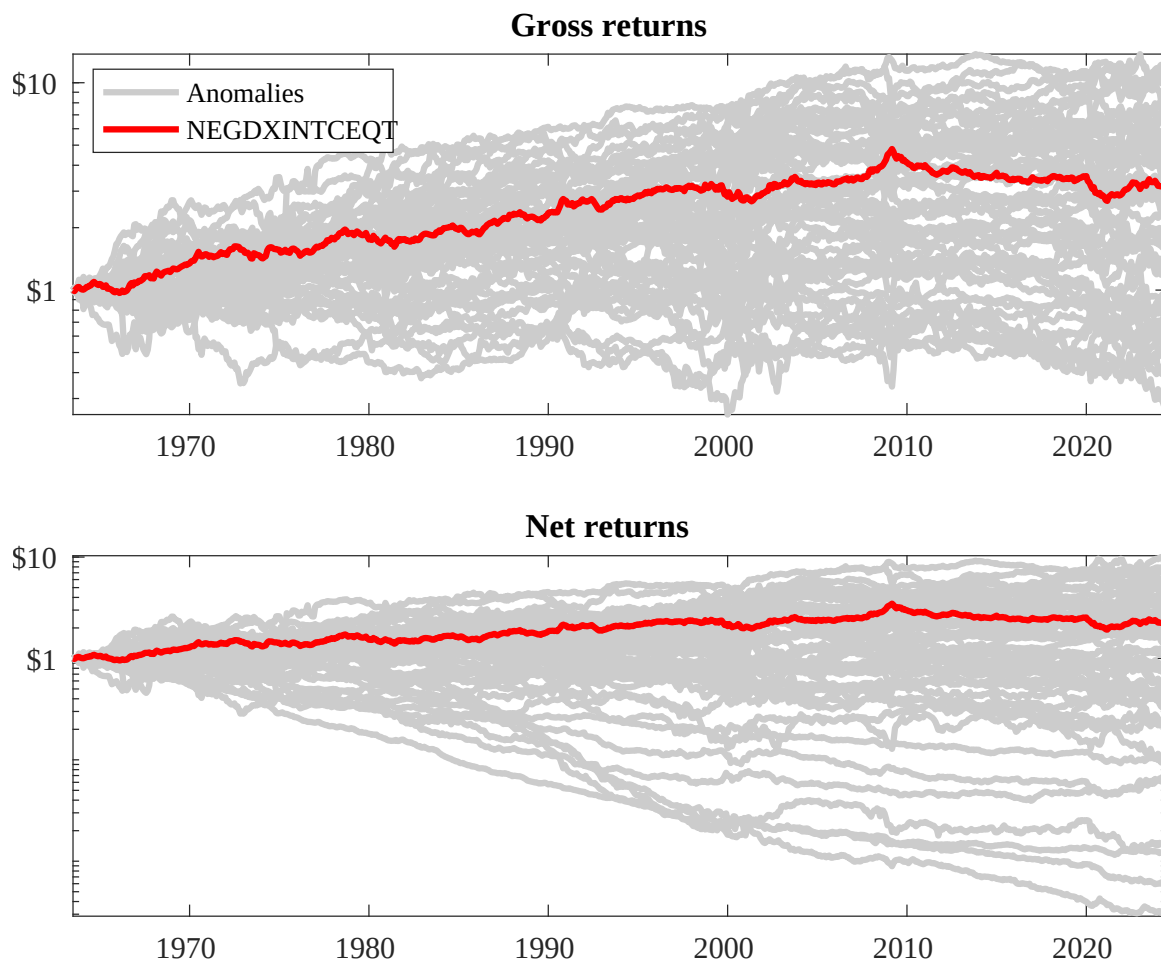
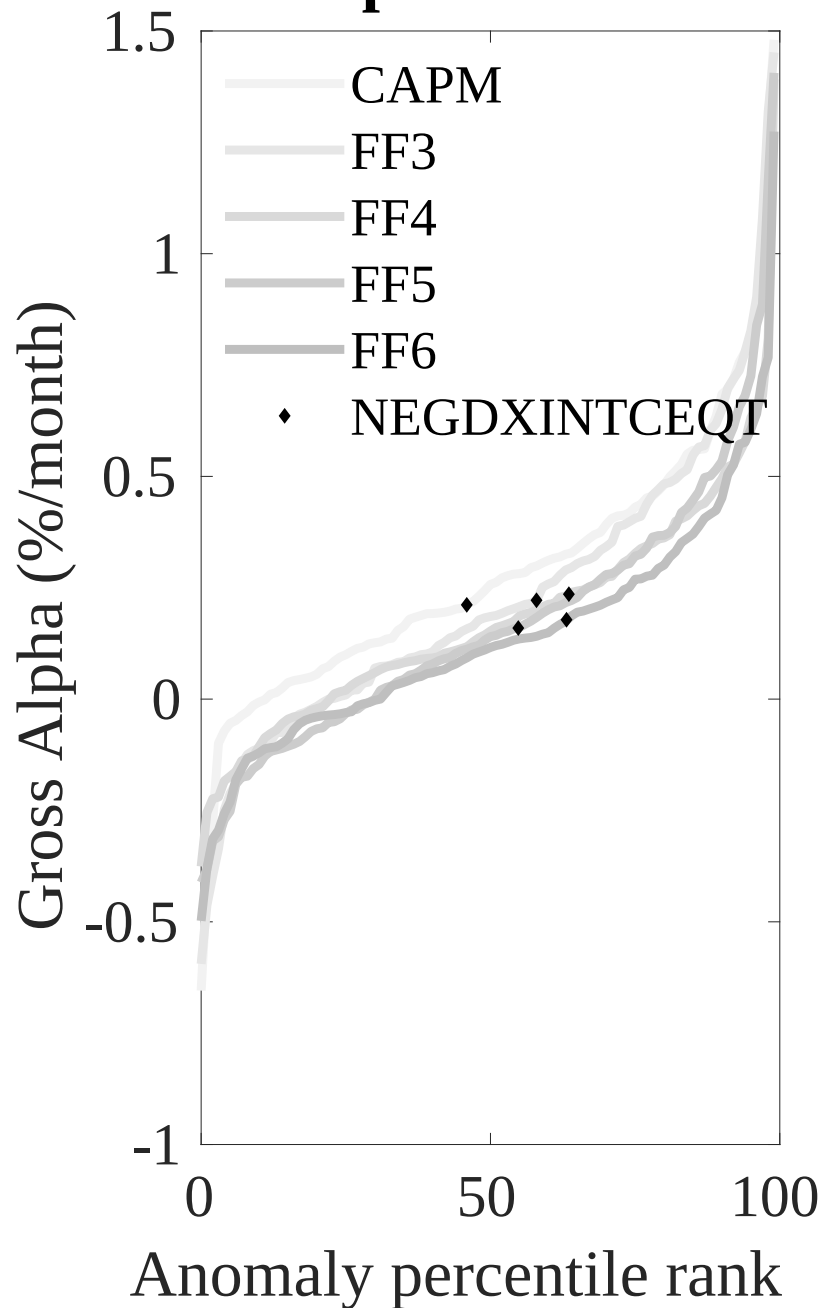


Figure 3: Dollar invested.

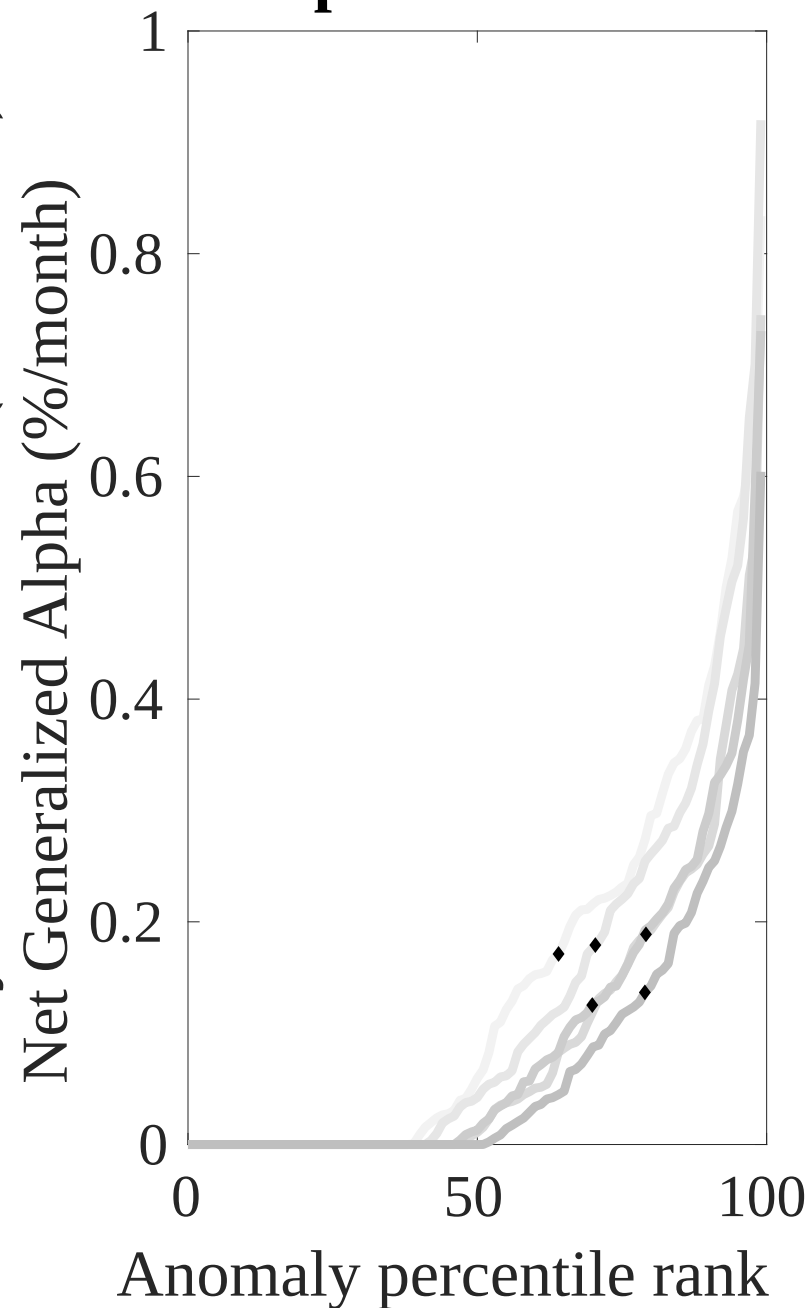
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the IBR trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



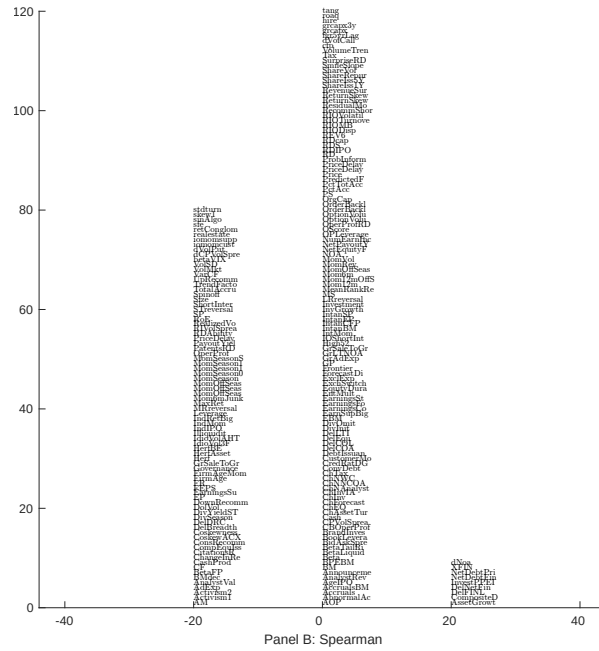
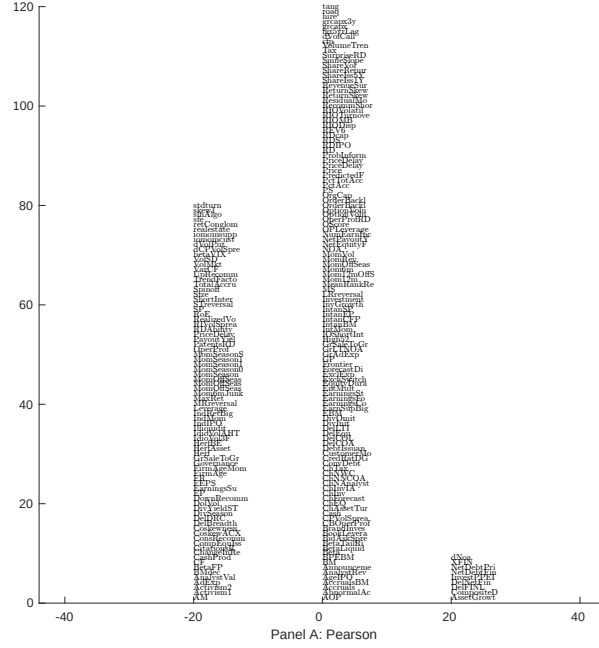


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 209 filtered anomaly signals with IBR. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

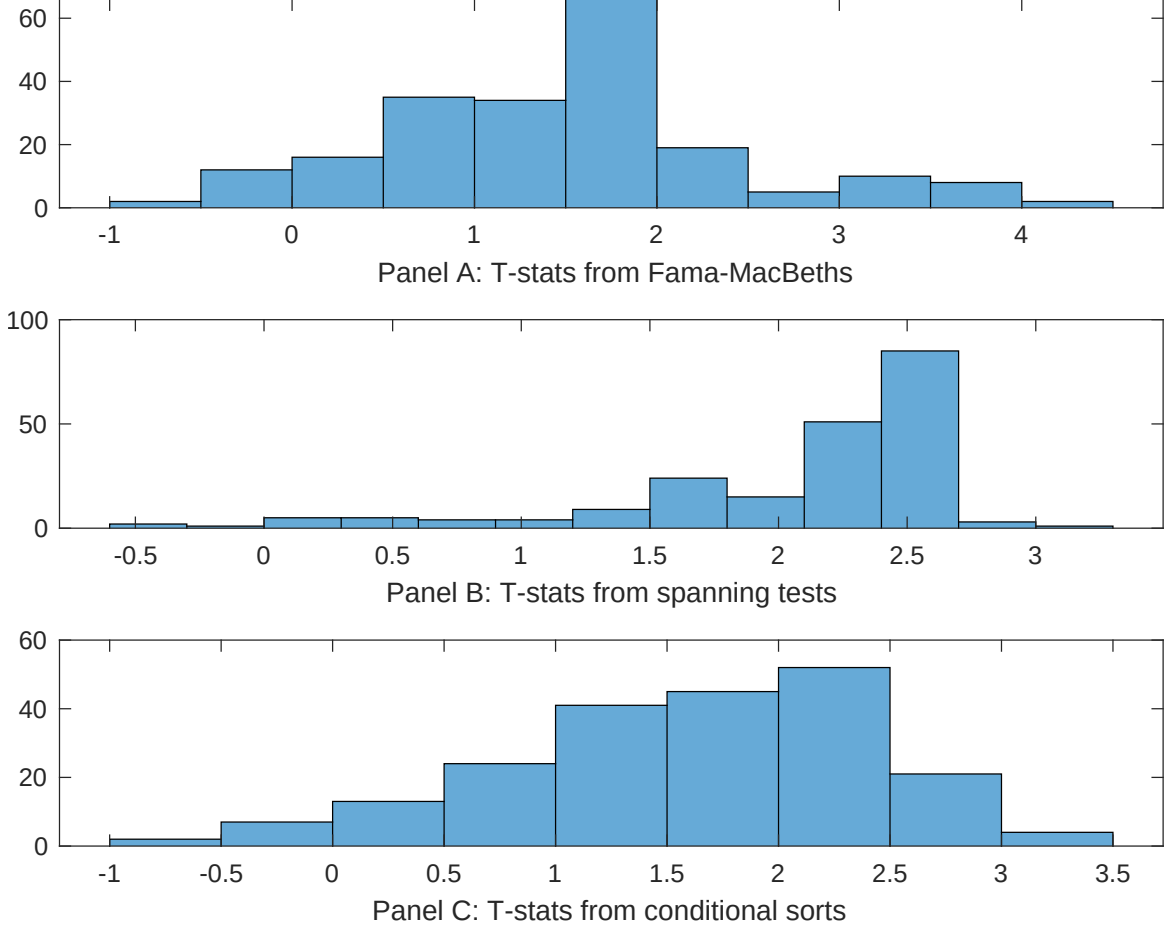


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of IBR conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{IBR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{IBR}IBR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{IBR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on IBR. Stocks are finally grouped into five IBR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted IBR trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on IBR. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{IBR}IBR_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in financial liabilities, Change in net financial assets, Net debt financing, Composite debt issuance, Net external financing, Revenue Growth Rank. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.12 [5.67]	0.12 [5.55]	0.13 [5.17]	0.13 [6.31]	0.13 [5.49]	0.10 [4.35]	0.11 [4.50]
IBR	-0.54 [-0.10]	0.19 [0.36]	0.70 [2.72]	0.11 [1.41]	0.61 [2.45]	0.45 [0.55]	-0.31 [-0.10]
Anomaly 1	0.18 [10.31]						0.62 [1.20]
Anomaly 2		0.96 [6.77]					0.30 [0.09]
Anomaly 3			0.20 [9.25]				-0.54 [-0.78]
Anomaly 4				0.95 [5.05]			0.56 [3.18]
Anomaly 5					0.21 [7.23]		0.14 [2.20]
Anomaly 6						0.20 [2.69]	0.13 [2.55]
# months	732	732	630	732	630	732	628
$\bar{R}^2(\%)$	0	0	0	0	1	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the IBR trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{IBR} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in financial liabilities, Change in net financial assets, Net debt financing, Composite debt issuance, Net external financing, Revenue Growth Rank. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.15 [2.20]	0.14 [1.95]	0.13 [1.72]	0.19 [2.71]	0.13 [1.64]	0.24 [3.44]	0.15 [1.99]
Anomaly 1	35.32 [8.84]						28.63 [4.80]
Anomaly 2		23.07 [6.81]					13.59 [3.40]
Anomaly 3			21.52 [5.03]				-10.76 [-1.92]
Anomaly 4				15.94 [4.71]			5.15 [1.38]
Anomaly 5					10.90 [2.75]		8.93 [2.32]
Anomaly 6						17.54 [5.93]	16.50 [5.15]
mkt	-6.12 [-3.72]	-6.60 [-3.94]	-5.88 [-3.25]	-6.08 [-3.55]	-4.36 [-2.27]	-5.93 [-3.50]	-2.40 [-1.34]
smb	0.03 [0.01]	5.01 [2.06]	1.89 [0.69]	1.22 [0.48]	7.46 [2.46]	1.90 [0.77]	2.54 [0.85]
hml	-9.93 [-3.11]	-12.77 [-3.94]	-12.04 [-3.47]	-14.87 [-4.48]	-11.96 [-3.39]	-15.58 [-4.73]	-13.54 [-4.03]
rmw	5.95 [1.85]	12.00 [3.64]	7.68 [2.16]	7.56 [2.27]	3.67 [0.86]	5.56 [1.66]	0.75 [0.19]
cma	4.43 [0.90]	21.93 [4.57]	11.29 [2.10]	13.73 [2.79]	10.21 [1.73]	4.45 [0.84]	-6.64 [-1.08]
umd	-5.51 [-3.32]	-3.17 [-1.91]	-4.24 [-2.34]	-3.43 [-2.02]	-3.06 [-1.68]	-3.08 [-1.84]	-5.47 [-3.19]
# months	732	732	630	732	630	732	630
$\bar{R}^2(\%)$	14	11	8	8	6	10	20

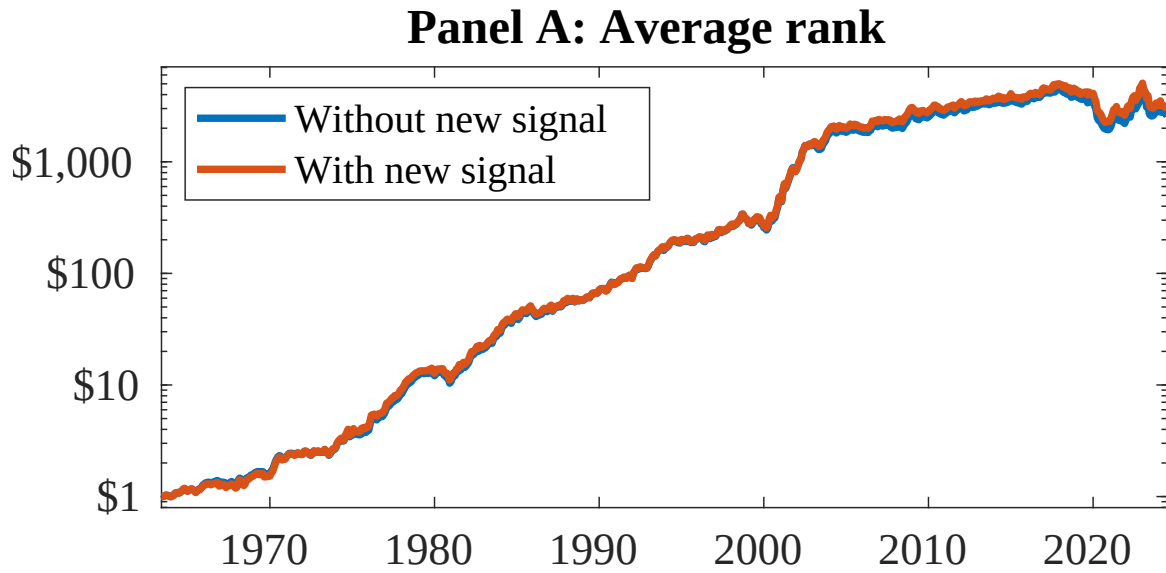


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as IBR. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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